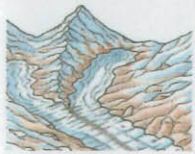


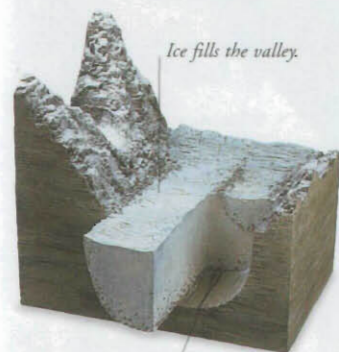
GLACIATION



THE SHAPING OF THE LANDSCAPE by ice is called glaciation. All over the world there are landscape features that were formed during past ice ages by glaciers, huge moving rivers of ice, and even bigger mounds of ice called ice sheets. In cold places, such as the polar regions, glaciers and ice sheets are still present, and glaciation still continues. The landscape created by ice is dramatic. Glaciers carve out deep, trough-like valleys, ice sheets pile up huge quantities of debris, and the icy conditions around can shatter rock into jagged peaks and knife-edge ridges.

How a glacier forms

Glaciers are created when layers of snow are compacted in icy mountain regions to form rivers of ice, which slowly creep downhill until they melt. The ice on the surface of the glacier cracks, forming deep crevasses, and both the surface and the underside of the glacier are covered with debris plucked away from the valley sides by the sheer weight of the passing ice.

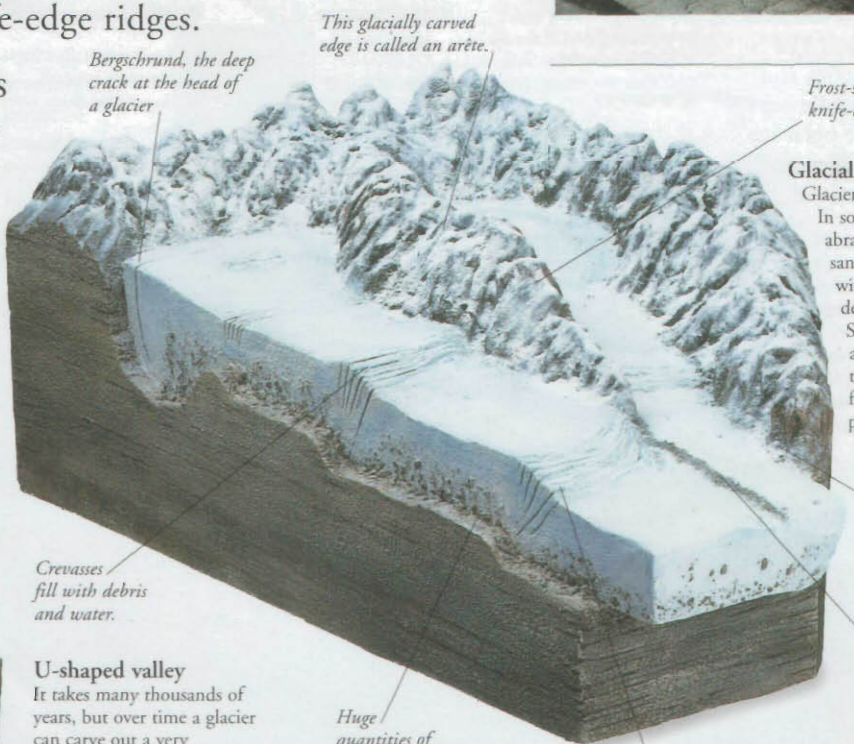


Debris is swept along beneath the glacier.

Traces of glaciation

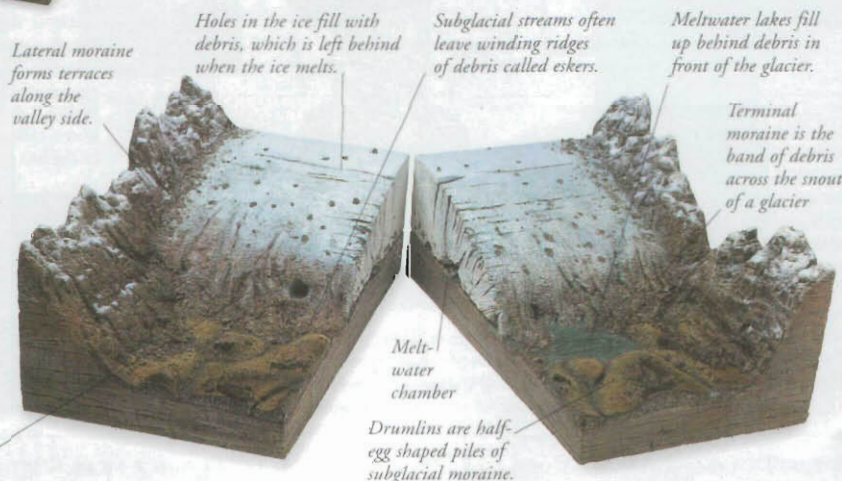
Glaciers carry huge quantities of debris, called moraine, which either fall on to the glacier from the mountains above or are swept away from the rock beneath. The moving ice pushes this debris into giant piles, or leaves it scattered over the landscape as the ice melts.

Lower end of the glacier



U-shaped valley

It takes many thousands of years, but over time a glacier can carve out a very distinctive, deep, U-shaped trough of a valley. If this reaches the coast and fills with seawater, it is called a fjord.



Valley glaciers

In high mountain ranges, such as the Alps and the Himalayas, glaciers form in valleys as snow slides from the peaks of the mountains. These are called alpine glaciers. Where these emerge from the mountains, they may cause piedmont glaciers, so called because they spread out in the shape of a foot.

Cirque, the deep hollow where the glacier begins.

Frost shatters rocky summits into jagged "horn peaks".



Valley glacier, Norway

Fjords

Fjords are steep-sided, narrow coastal inlets, formed where glaciers have ground out deep valleys along existing riverbeds. When the ice melted, the valleys were flooded as the sea-level rose. The coast of Norway has many fjords.

Frost-shattered rock falls on to the ice as lateral moraine along the side of the glacier.

Medial moraine - a band of moraine formed as two glaciers flow together.

Ice fall, where the ice flows over a step in the valley floor.

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